

MA 574: An Introduction to Probabilistic Machine Learning

Course Code	MA 574
Course Name	An Introduction to Probabilistic Machine Learning
L-T-P-C	3-0-2-4
Prerequisites	Probability and Statistics, Linear Algebra, and Optimizations
Intended for	B. Tech/ MS/PhD/MSc
Distribution	Elective
Mutual Exclusion	DS 413, CS 669
Approval	59th BoA (Apr 2025)

Course Contents

- **Foundations of Probabilistic Reasoning:** Review of probability distributions and statistical inference. Maximum Likelihood Estimation (MLE), Bayesian inference, and Maximum a Posteriori (MAP) estimation. Fundamental information-theoretic concepts: entropy, conditional entropy, mutual information, and the Kullback-Leibler (KL) divergence. [6 Lectures]
- **Generative Models for Supervised Learning:** Generative approaches to classification: Naive Bayes and Gaussian Discriminant Analysis (GDA). Structured probabilistic models, Bayesian networks. Parameter estimation and inference techniques. [8 Lectures]
- **Probabilistic Discriminative Models:** Logistic regression and its theoretical foundation. Overfitting and generalization: regularization techniques. Introduction to the perceptron algorithm and comparison with probabilistic models. Overview of Bayesian logistic regression. [8 Lectures]
- **Ensemble Learning and Non-linear Extensions:** Non-linear and ensemble-based methods: decision trees, entropy, and information gain criteria. Ensemble methods: bagging, random forests, boosting, AdaBoost, and Gradient Boosting algorithms. Probabilistic interpretation of ensembles and the bias-variance trade-off. Introduction to Gaussian Processes (GPs) and how Bayesian non-parametric models define priors over functions. [10 Lectures]
- **Latent Variable Models and Unsupervised Learning:** Modeling unobserved structure: Expectation-Maximization (EM) algorithm, Gaussian Mixture Models (GMMs), and the method of moments. Unsupervised learning methods: k-means clustering and spectral clustering. Dimensionality reduction for high-dimensional data: probabilistic PCA and visualization techniques like t-SNE and UMAP. [10 Lectures]

Laboratory Modules

The lab component of this course focuses on the practical application of concepts via hands-on implementation with actual datasets. It aims to enhance students' understanding of concepts and strengthen their ability to use statistical and machine approaches and evaluate results effectively.

Text Books

1. 1. Murphy, J. P., **Machine Learning: A Probabilistic Perspective**, The MIT Press, 2012.
2. 2. Mitchell, T. M., **Machine Learning**, McGraw Hill, 2017.

References

1. Shai, S. S., and Shai, B. D., **Understanding Machine Learning: From Theory to Algorithms**, Cambridge University Press, 2014.
2. Duda, R. O., Hart, P. E., and Stork, D. G., **Pattern Classification**, 2nd Edition, Wiley-Interscience, 2000.
3. Tan, P. N., Steinbach, M., and Kumar, V., **Introduction to Data Mining**, Pearson, 2014.
4. Bishop, C. M., **Pattern Recognition and Machine Learning**, Springer, 2006.